

STAT482 Student Instruction Handout

Case: Building a Mixed-Data Business Analytics Dashboard with ChatGPT and R Shiny

Purpose. Work in teams as analytics consultants. Use ChatGPT to support idea generation, coding, and explanation drafting, but verify every result before submission.

1. Case scenario

Client need. A regional retailer wants an interactive dashboard that reads an Excel worksheet containing both quantitative and qualitative variables and returns descriptive summaries on request.

- Quantitative examples: order value, discount rate, wait time, delivery days, satisfaction score, and NPS score.
- Qualitative examples: region, channel, product category, customer segment, loyalty tier, complaint theme, and manager notes.
- The dashboard should help a manager inspect the data quickly and support simple business decisions.

2. Team tasks

1. Audit the Excel workbook and classify each variable correctly.
2. Clean missing or inconsistent values and document your decisions.
3. Use ChatGPT to brainstorm dashboard features and write or debug R code.
4. Build a working R Shiny dashboard that uploads the workbook and summarizes the selected worksheet.
5. Interpret the outputs as business insights and note any data-quality limits.

3. Required deliverables

- **Brief report:** 4-6 pages covering the problem, data preparation, dashboard design, descriptive findings, validation steps, and recommendations.
- **Oral presentation:** 8-10 minutes with a dashboard demonstration, major results, and a short reflection on ChatGPT use.

4. Suggested dashboard features

- Excel upload and worksheet selection.
- Overall numeric summary table with n, missing, mean, median, standard deviation, minimum, and maximum.
- Frequency table for a selected qualitative variable.
- Grouped summary of a numeric variable by a category such as region or channel.

- A plot for quick exploration, such as a histogram or bar chart.
- A searchable preview of the raw data.

5. Recommended workflow

6. Business framing: identify the users, decisions, and questions the dashboard should answer.
7. Data preparation: inspect variable types, missing values, and inconsistent labels.
8. Dashboard development: use R Shiny, readxl, dplyr, ggplot2, and DT to build the interface and calculations.
9. Validation: compare selected outputs with manual checks in Excel or simple R summaries.
10. Interpretation: explain what the descriptive results mean for a manager.

6. Responsible use of ChatGPT

Important. ChatGPT may help you classify variables, outline a Shiny interface, draft code, or debug problems. It may also make mistakes. You are responsible for checking all statistics, all code, and all interpretations before submission.

- Useful prompt: Help me classify these Excel variables as quantitative, qualitative, ordinal, date, or text.
- Useful prompt: Suggest a simple R Shiny layout for mixed business data with summary statistics on request.
- Useful prompt: Why does my grouped summary fail when the user selects a variable in Shiny?
- Useful prompt: What common mistakes should I check in a descriptive statistics dashboard?

7. Submission checklist

- The app runs and reads the sample workbook.
- Numeric and categorical summaries are clearly labeled.
- Missing values are handled or explained.
- ChatGPT use is documented and verified.
- The presentation includes a dashboard demonstration.
- All files are organized clearly for submission.

8. Included files in this package

- STAT482_sample_dashboard_dataset.xlsx
- app.R
- STAT482_case_report_rubric.docx
- STAT482_presentation_rubric.docx